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Estimating multivariate ecological variables at high spatial resolution using a cost-effective matching algorithm

Rachel R. Renne[1,2] | **Daniel R. Schlaepfer**[1,2,3] | **Kyle A. Palmquist**[4] | **William K. Lauenroth**[1] | **John B. Bradford**[2]

[1]Yale School of the Environment, Yale University, New Haven, Connecticut, USA

[2]U.S. Geological Survey, Southwest Biological Science Center, Flagstaff, Arizona, USA

[3]Center for Adaptable Western Landscapes, Northern Arizona University, Flagstaff, Arizona, USA

[4]Department of Biological Sciences, Marshall University, Huntington, West Virginia, USA

Correspondence
Rachel R. Renne
Email: rachel.renne@yale.edu

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Abstract

Simulation models are valuable tools for estimating ecosystem response to environmental conditions and are particularly relevant for investigating climate change impacts. However, because of high computational requirements, models are often applied over a coarse grid of points or for representative locations. Spatial interpolation of model output can be necessary to guide decision-making, yet interpolation is not straightforward because the interpolated values must maintain the covariance structure among variables. We present methods for two key steps for utilizing limited simulations to generate detailed maps of multivariate and time series output. First, we present a method to select an optimal set of simulation sites that maximize the area represented for a given number of sites. Then, we introduce a multivariate matching approach to interpolate simulation results to detailed maps for the represented area. This approach links simulation output to environmentally analogous matched sites according to user-defined criteria. We demonstrate the methods with case studies using output from (1) an individual-based plant simulation model to illustrate site selection, and (2) an ecosystem water balance simulation model to illustrate interpolation. For the site selection case study, we identified 200 simulation sites that represented 96% of a large study area (1.12×10^6 km^2) at a ~1-km resolution. For the interpolation case study, we generated ~1-km resolution maps across 4.38×10^6 km^2 of drylands in North America from a 10 × 10 km grid of simulated sites. Estimates of interpolation errors using cross validation were low (<10% of the range of each variable). Our point selection and interpolation methods, which are available as an easy-to-use R package, provide a means of cost-effectively generating detailed maps of expensive, complex simulation output (e.g., multivariate and time series) at scales relevant for local conservation planning. Our methods are flexible and allow the user to identify relevant matching criteria to balance interpolation uncertainty with areal coverage to enhance inference and decision-making at management-relevant scales across large areas.

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KEYWORDS
ecohydrology, ecological modeling, multivariate interpolation, multivariate matching, sagebrush, sampling design, time series interpolation

INTRODUCTION

Ecosystem structure and function are influenced by factors that vary across space and time: climate exerts important control at broad scales and acts in concert with local and site-level heterogeneity in topography, edaphic conditions, and disturbance history, among other factors. As a result, estimating ecosystem structure and function at fine scales over large areas is challenging, particularly in heterogeneous landscapes (Estes et al., 2018; Fritsch et al., 2020). Direct observation is typically limited by time and resources to a finite number of locations and observations are often only occasional or episodic. By contrast, mechanistic simulation modeling, built upon relationships derived from empirical data, can be a useful tool for assessing ecological processes over a wide range of conditions (Grimm & Berger, 2016; Pennekamp et al., 2017; Radchuk et al., 2019). For example, such models can be used for projecting the potential impacts of multiple climate change scenarios and their interactions with disturbance on ecosystem structure and function (Jeltsch et al., 2008).

Although simulation models are useful for representing the impact of a wide range of conditions, estimating complex, data-intensive processes like plant community dynamics or ecosystem water balance at management-relevant scales over large geographic areas is often limited by computational requirements. For example, estimating ecosystem structure and function at 1-km resolution for a region of interest could mean simulating hundreds to millions of sites. In addition, the number of necessary model simulations quickly compounds when assessing a suite of future climate scenarios, disturbance and management scenarios (e.g., fire, grazing), and environmental conditions (e.g., soil properties) even for a relatively small number of sites. As a result, models are often run at relatively low resolution (Notaro et al., 2012; Reeves et al., 2017; Schlaepfer et al., 2017) or for representative or sparse, regularly-spaced points (Bradford et al., 2019; Palmquist et al., 2016b; Renwick et al., 2017). The site-specific nature of these results can be difficult to use for decision-making at local scales (Zimmer et al., 2021), particularly in regions with highly heterogeneous environmental conditions (i.e., highly variable topography and soils).

Efforts to downscale low-resolution model results or interpolate representative points to create continuous maps usually focus on one variable, such as environmental suitability represented by species distribution models (Keil et al., 2013) or soil moisture (Kearney et al., 2020). However, these efforts are not designed to appropriately interpolate multivariate output (such as output from mechanistic simulation models) consisting of a set of related variables whose relationships must be maintained. For multivariate datasets, interpolating variables individually can create incoherence because it ignores covariance among output variables (e.g., how the relative abundance of one plant functional type is related to the abundance of other types).

Available approaches for interpolating multivariate simulation output include statistical emulators, geostatistical methods, and analog methods. Statistical emulators use machine learning techniques to predict simulation output variables from a set of simulation runs for conditions that have not been simulated when models are computationally expensive (O'Hagan, 2006). For example, emulators have been used on spectral output from computationally expensive radiative transfer models that simulate how light interacts with vegetation, atmosphere, and water for remote sensing applications (Vicent et al., 2018). Geostatistical methods such as geographically weighted regression and co-kriging use geographic proximity, sometimes in combination with other predictors, to estimate a variable between locations with known values (Brunsdon et al., 1996; Myers, 1982). For example, efforts to create high-resolution, multivariate and time series weather datasets via interpolation have developed equations based on relationships between topography, radiation, and multiple weather variables recorded at meteorological stations (e.g., Thornton et al., 1997). Although these methods may draw upon multiple covariates for interpolation, they interpolate one variable at a time and are inadequate for interpolating high-dimensional output from simulation models. Analog methods are widely used in the fields of meteorology and climatology to predict local weather in near (in hours, days, and weeks) and distant (in years and decades) time frames (Gangopadhyay et al., 2005; Horton et al., 2017; Yates et al., 2003). Analog methods use multivariate matching to identify analogs between current and past large-scale atmospheric characteristics and predict weather by assigning past local weather conditions from the best match or one of a series of best matches as the predicted weather conditions (Lorenz, 1969; Zorita & Von Storch, 1999).

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In our case, we wanted to interpolate geographically sparse, high-dimensional and time series simulation output across heterogeneous terrain using a cost-effective and flexible method that maintained multivariate relationships among output variables. Developing a statistical emulator for this problem would have been computationally costly and current geostatistical methods do not handle interpolation of high-dimensional data. Thus, we drew upon the concept of weather analogs to develop a nearest neighbor interpolation method that matches the nonsimulated sites with their most similar environmental analog from among simulated sites, where similarity between neighbors is measured in environmental space instead of geographic space. The method assigns simulation output from simulated sites to their matched nonsimulated sites to create continuous maps of complete simulation output. Our technique for identifying analogs allows the user to explicitly specify what constitutes an environmentally analogous site using domain expertise and provides methods to evaluate matching and to quantify the uncertainty arising from the method. Criteria for identifying environmental analogs can be fine-tuned to balance uncertainty with the areal coverage of the interpolated variables.

First, we present a method for selecting a subset of sites to be used to generate data for interpolation. This method identifies an optimal set of sites from within an area of interest that maximize the proportion of the area that can be reasonably represented by those sites while minimizing the number of sites needed given the user selected matching quality (*subset cells*, Figure 1). We demonstrate the site selection method by identifying optimal sites for simulation of dryland plant community dynamics to represent sagebrush ecosystems across western North America. Then, we present a method for interpolating simulation output from a defined set of sites to produce continuous maps using multivariate matching (*output variables*, Figure 1). We demonstrate interpolation using results from an ecosystem water balance model applied in a coarse grid of points to produce continuous, high-resolution maps of water balance estimates across North American drylands. Our methods are highly flexible, and we have developed our code from this project into an R package to make our methods available for other applications (Renne, 2021).

MATERIALS AND METHODS

We propose algorithms based on weighted multivariate matching to (1) select an optimal set of sites for simulation and (2) spatially interpolate simulation output to continuous maps at fine spatial scales (Figure 1). These algorithms can be used together or individually. Identifying *matching variables* and *matching criteria* are required for both algorithms (Figure 1) and are described in the site selection section below. Unless otherwise noted, we conducted all calculations and analyses in R v4.0.3 (R Core Team, 2020). The code and functions for applying our methods are freely available as an R package (https://zenodo.org/records/5784247; Renne, 2021).

Method 1: Selecting optimal sites for simulation and/or sampling

Step 1: Identify variables for matching

Our site selection method uses environmental variables (e.g., climate and soil variables) supplied as raster data from an area of interest as matching variables to identify groups of raster cells that are environmentally and statistically similar and to identify one representative cell from each group as one of the subset cells to be simulated (Figure 1). Cells are grouped using a measure of similarity that incorporates ecosystem- and process-relevant criteria defined by the user (see *Step 2*). If sites are required for simulation and/or field sampling, they can be identified as the geographic centroids of the cell representing each group. Our method for selecting subset cells maximizes the proportion of the study area that can be reasonably represented by these cells given the specified criteria. The data used to measure similarity for matching will depend on the goals of the site selection effort and should be based on knowledge of the system and processes in question. For example, we used environmental variables that influence plant community structure (mean annual precipitation and temperature, and a measure of precipitation seasonality) to select sites for simulations of plant community dynamics in North American temperate drylands. This approach is flexible and can be generalized to other systems using matching variables and matching criteria that are relevant to those systems. Importantly, the resolution of the interpolated output variables is determined by the resolution of the matching variables.

Step 2: Define criteria for matching variables

Matching variables are often measured in different units and/or exhibit large differences in magnitude across a study area. To prevent variables measured in large units from outweighing those measured in smaller units during the matching process, multivariate methods often standardize variables. Our methods identify groups of cells that are environmentally similar with the

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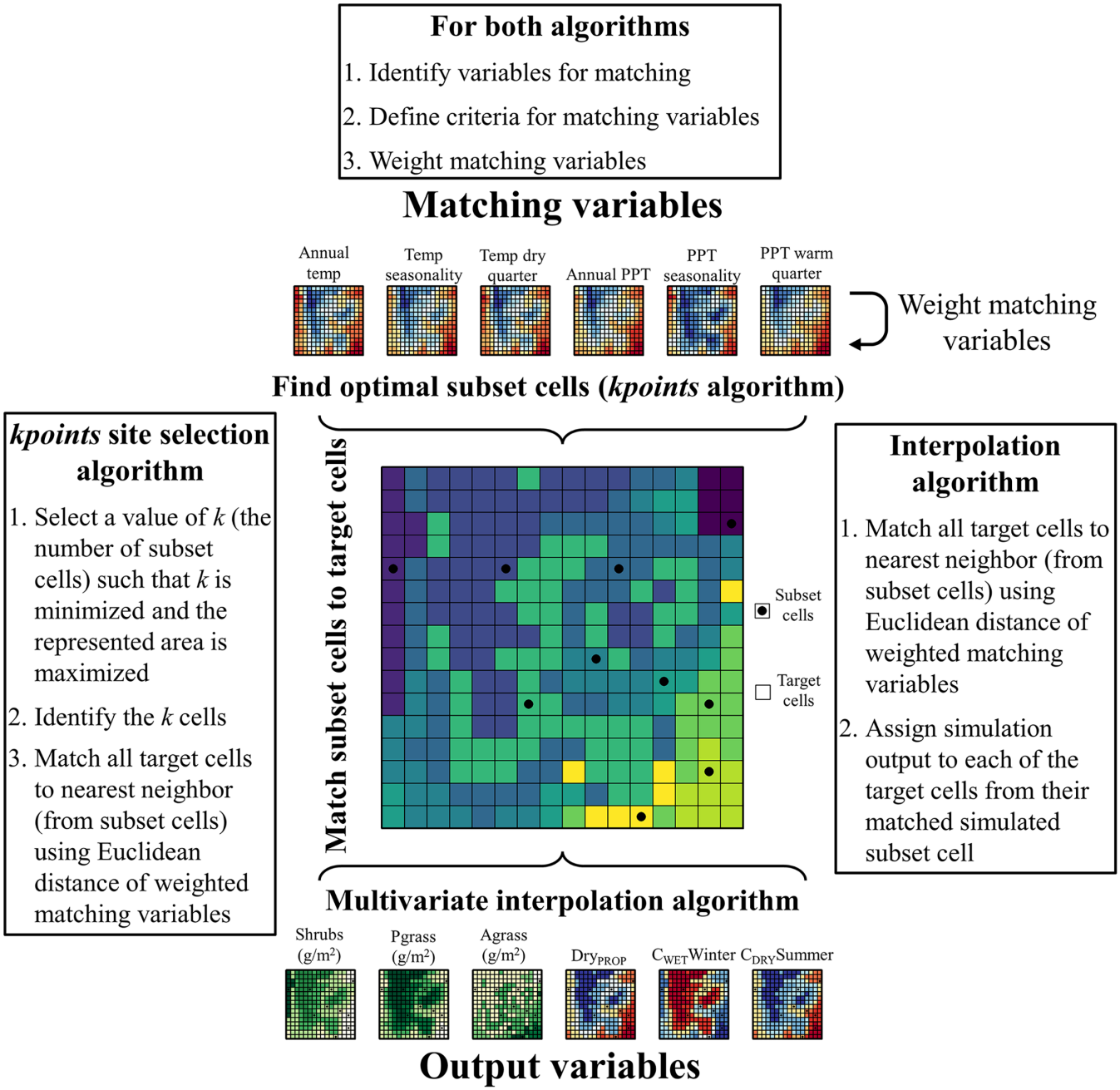


FIGURE 1 Conceptual diagram and overview of the site selection and interpolation methods. First, *matching variables* are selected and weighted with user-defined matching criteria. For site selection, the *kpoints* algorithm is used to determine an optimal set of *k subset cells* for use in field sampling or simulation. All *target cells* are matched to one of the subset cells (matches are denoted by color in the central panel). Point results (e.g., simulation output) for those cells are interpolated across the study area using multivariate matching to produce continuous, high-resolution maps of output, such as plant functional type biomass or ecohydrological variables. Temperature seasonality is the SD of average monthly temperatures and precipitation seasonality is the CV of average monthly precipitation. Agrass, annual grass; $CDRY_{SUMMER}$, number of consecutive days with all MCS layers dry during the summer; $CWET_{WINTER}$, number of consecutive days with all MCS layers wet during the winter; DRY_{PROP}, proportion of days that all layers within the moisture control section (MCS: soil layers with depth ranging from 10 to 30 cm for fine textures to 30–90 cm for coarse textures; Soil Survey Staff, 2014) of the soil are dry when soil temperature at 50 cm is >5°C; Pgrass, perennial grass; PPT, precipitation; Temp, temperature.

assumption that environmentally similar sites will have similar simulation output. The definition of "environmentally similar" is determined using the matching criterion for each matching variable. Matching criteria are defined as the maximum acceptable difference between subset cells and their matches (target cells) for

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each matching variable (Figure 1). To assess similarity between sites, each matching variable is standardized by dividing it by its designated matching criterion so that the units of all variables are scaled to be equivalent in terms of "environmental similarity." Thus, similarity between cells is calculated as the Euclidean distance of these scaled (i.e., weighted) matching variables. Cells that are within one unit of this weighted Euclidean distance from each other will be within the user-defined criteria for all matching variables.

Defining appropriate matching criteria is critical to both site selection and interpolation and should be based on domain expertise. Conceptually, matching criteria are meant to rescale matching variables such that one unit change in any rescaled variable would result in a similar level of dissimilarity between sites. We recommend carefully reviewing and reporting evaluation metrics (see *Evaluating matching and estimating matching error*) to ensure that the uncertainties in represented area and interpolation are acceptable for the application and clearly communicated along with the data. It may be useful to start with a tentative set of matching criteria, perform matching, and then evaluate matching using evaluation metrics 1–3 (which assess environmental distance between sites, within-group homogeneity for matches, and geographic distances between target and matched subset cells). Matching criteria can be adjusted based on these results to balance matching uncertainty with the areal coverage of the methods (Figure 2a). A final check of matching criteria is the estimated interpolation error from evaluation metric 4 (leave-one-out cross validation of interpolation results). If errors are large, reducing matching criteria will likely reduce estimated matching error. Importantly, this will also reduce the represented area and care must be taken to weigh the cost of this uncertainty versus the area that can be reasonably represented by the target sites (Figure 2a).

Overview of the site selection algorithm

To identify k representative sites within an area of interest (subset cells, Figure 1), we developed an algorithm, *kpoints* (see the "kpoints()" function in our R package), based on the unsupervised classification method kmeans (MacQueen, 1967). First, k cells (where k is the number of sites to be simulated and/or sampled in the field) are randomly selected from all cells in the study area (target cells). Target cells are assigned to one of the k subset cells by determining their nearest neighbor from

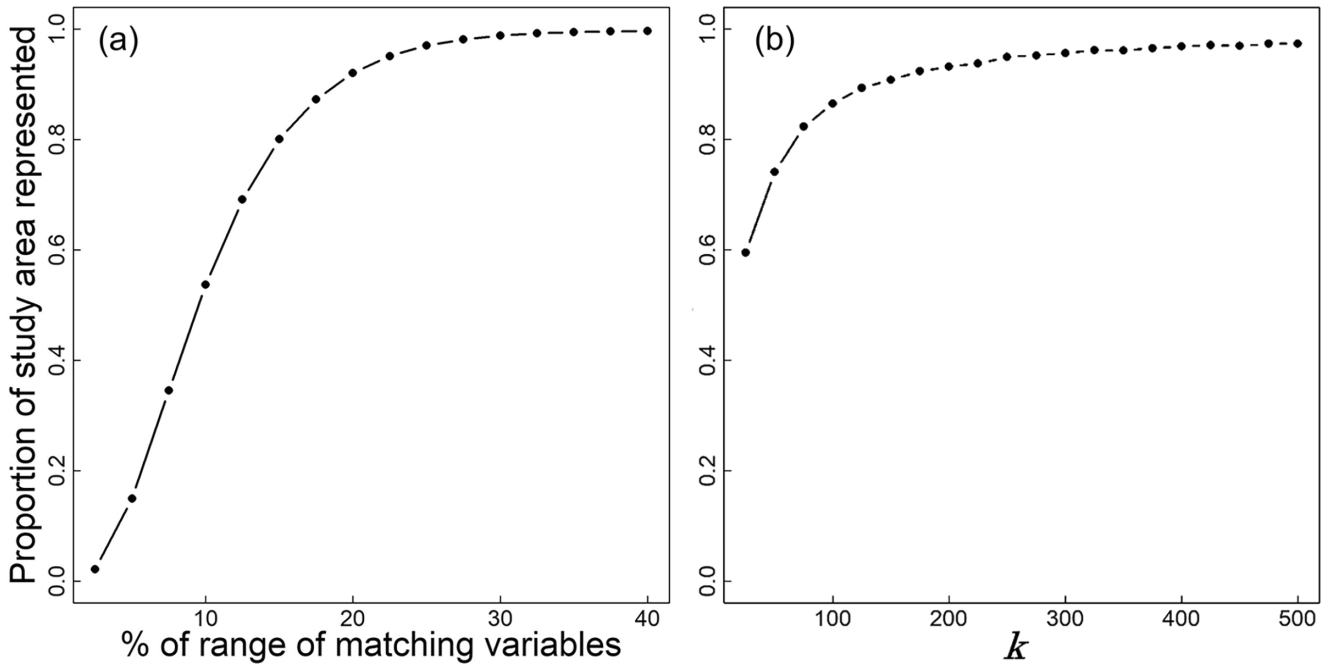


FIGURE 2 Proportion of the study area in case study 1 represented by 200 subset cells under a range of different matching criteria (a) and by different numbers of k subset cells selected using the *kpoints* algorithm with fixed matching criteria (b). Matching criteria in (a) are calculated as different percentages of the range of each matching variable. Area represented increased with matching criteria (left to right along x-axis in (a)), but uncertainty also increased because matched subset and target cells could have larger differences in matching variables. In (b), increases in coverage began to level off around $k = 200$ and we selected this as the final value for k. Represented area is defined as the area of cells with Euclidean distance ≤1 for the weighed matching variables.

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among the subset cells using Euclidean distance of the weighted matching variables, resulting in k groups of target cells. The centroids (multivariate mean) of the k groups of target cells are calculated and the cell that is the nearest neighbor in multivariate space to the centroid of each group becomes one of an updated set of subset cells. In the next iteration, the rest of the target cells in the study area are matched with one of the updated k subset cells and the process continues through additional iterations until a stopping criterion is met: either a predefined number of iterations are completed ("iter" argument in "kpoints()" function) or the change in area represented is at or below a designated threshold ("min_area" argument in the "kpoints()" function) for five consecutive iterations.

Importantly, the final iteration will not always be the iteration that represents the largest proportion of the study area, so the algorithm stores the selection of k subset cells and the area represented for all iterations. Once a stopping criterion is met, the k subset cells for the iteration that represented the largest proportion of the study area are saved as the best solution. Stopping criteria are designed to stop the algorithm when an optimal solution has been found (little to no gain in represented area for additional iterations) and to prevent the algorithm from running indefinitely by setting the maximum number of iterations ("iter"). Because restricting the number of iterations ("iter") may cause the algorithm to stop before an optimal solution is found, values for "iter" and "min_area" must be chosen to ensure that the change in area represented by each iteration levels out before stopping (set "verify_stop" to TRUE in "kpoints()" function; Appendix S1). We provide a detailed, step-by-step example of how to use the "kpoints()" function in the *Selecting points with rMultivariateMatching* vignette included with our R package (Renne, 2021).

Step 3: Identify the number of sites (k)

The number of sites (k) to be simulated and/or sampled in the field can be fine-tuned by examining the relationship between the number of sites and the area represented by those sites. This is achieved by finding solutions for the *kpoints* algorithm for a range of possible values for k that includes the maximum number of sites that can feasibly be sampled or simulated. The area represented will increase with the number of sites, but gains may level off as the number of sites increases (e.g., Figure 2b). Users can choose an optimal number of sites by weighing computational costs against the benefits of representing more area for different numbers of sites. The "kpoints()" function can be run for a range of possible values for k using the "klist" argument. Displaying the results with the "plotcoverage()" function can be useful for visualizing the relationship between areal coverage and k when determining the optimal number of points. Importantly, matching criteria can also be evaluated and fine-tuned during this step by comparing the change in represented area when the matching criteria are adjusted for a fixed number of sites (see the "choose_criteria()" function and the *Selecting points with rMultivariateMatching* vignette in our R package). Importantly, relaxing matching criteria will increase matching uncertainty and interpolation errors and care should be taken to balance the benefits of increasing the areal coverage with the cost of increasing uncertainty (Figure 2a).

Step 4: Identify the subset cells

Like kmeans, the solution for *kpoints* is sensitive to the initial subset of randomly selected subset cells (Celebi et al., 2013) and the algorithm should be run multiple times with different initial random selections of cells. The number of runs is designated by the "n_starts" argument in the "kpoints()" function. When "n_starts" is greater than one, the function stores the subset cells and the proportion of the area represented for each of the runs. At the end, it compares the proportion of area represented across runs and returns the solution from the run that represented the largest proportion of the study area as the final solution. When running the "kpoints()" function with different values of k to determine the optimal number of sites for a given set of matching criteria (see *Step 3: Identify the number of sites (k)*), a small number of runs (e.g., "n_starts" = 10) can be used. Running the "kpoints()" function with only a small number of runs may underestimate the potential represented area for a given number of sites, but should be adequate for efficiently determining the optimal number of sites. Once the number of sites is determined, a much larger number of runs (e.g., "n_starts" = 100) will help maximize the area represented by the final selection of sites. This process is detailed in the *Selecting points with rMultivariteMatching* vignette in our R package.

Case study for method 1: Selecting optimal sites for simulation and/or sampling

Our goal was to apply the *kpoints* algorithm to find an optimal subset of sites to simulate using STEPWAT2, an individual-based, gap dynamics plant simulation model (Palmquist, Bradford, et al., 2018; Palmquist, Schlaepfer, et al., 2018), to project the impacts of climate change, wildfire, and livestock grazing on big sagebrush (*Artemisia tridentata* Nutt.) plant communities across

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their extent in the western United States. We defined an area of interest of approximately 1.19×10^6 km^2 defined by 1.91×10^6 30-arcsecond (~1 km) cells (detailed in Appendix S2). It was imperative to conduct simulations for a relatively small number of subset cells because the experimental design included current conditions, 52 future climate scenarios, 4 grazing treatments, and 4 fire treatments, resulting in 848 unique combinations for each site.

In STEPWAT2, soil water that is available for transpiration determines resource availability and ultimately plant biomass and is largely governed by climate and soil variables (Palmquist, Bradford, et al., 2018). We controlled for soil using a single representative soil type for simulations (median soil type across big sagebrush ecosystems, Palmquist et al., 2021: Appendix S4), and we selected six key climate variables (Table 1) that represent the major climatic drivers of plant community structure in big sagebrush ecosystems (Paruelo & Lauenroth, 1996; Renne et al., 2019) as matching variables for site selection. We set matching criteria for differences between subset and target cells to 10% of the range of each variable (Table 1). We used DayMet (Thornton et al., 2018) data to derive 30-year (1981–2010) normal monthly precipitation and temperature for every cell within our study area using Google Earth Engine (Gorelick et al., 2017) and calculated matching variables from these values using the "biovars" function in the "dismo" R package (Hijmans et al., 2017).

We ran the *kpoints* algorithm for all cells within our study area, with the maximum number of iterations ("iter") set to 50 and the stopping criterion for change in represented area ("min_area") set to 50 km^2. We determined the optimal number of sites (k) by finding solutions from 25 to 500 in steps of 25 using 10 runs ("n_starts" = 10) of the "kpoints()" function for each value of k. After we determined the optimal number of sites, we obtained our final selection of subset cells by running the "kpoints()" function for 100 runs ("n_starts" = 100).

Method 2: Interpolation

Our method for generating high-resolution maps of complex simulation output can be used for any set of simulated sites (representing subset cells, Figure 1). These sites may have been selected by any approach, including using the method described above, defined by establishing a regular grid of points over the study region (as in the case study below), or defined by the locations of long-term research sites. Our method uses multivariate matching to interpolate results from individual sites to a broader study region by assigning simulation results from each simulated site to all cells within an area of interest that can be considered "environmental analogs" according to the matching criteria. That is, all cells in the study area are matched to an analogous simulated site; then, simulation output is assigned to all cells from their simulated analogs to generate continuous maps of all simulation output (Figure 1). We provide three vignettes with step-by-step examples of interpolation in our R package. First, we demonstrate interpolation using simulation output from points selected using the "kpoints()" function (*Matching and interpolation with kpoints*). We also demonstrate interpolation using simulation output from sites selected in an even grid across the study area and matched to target cells using (1) a simple, one-step matching approach (*Matching and interpolation without kpoints*) or (2) a two-step matching approach as described below (*Two-step matching and interpolation without kpoints*). We provide a validation of this interpolation method in Appendix S3.

Case study for method 2: Interpolation

Our goal was to generate 30-arcsecond maps from coarse grid of simulation output from SOILWAT2, a process-based ecohydrological simulation model (Schlaepfer et al., 2012; Schlaepfer & Andrews, 2018; Schlaepfer & Murphy, 2018).

TABLE 1 Summary of six matching variables for the site selection case study with matching criteria (maximum difference between subset and target cells), defined as 10% of the range of each variable.

Metric	Mean annual temperature (°C)	Mean annual precipitation (mm)	Temperature seasonality	Precipitation seasonality	Mean temperature of driest quarter (°C)	Mean precipitation of warmest quarter (mm)
Mean	7.39	395	874	44.2	9.10	81.5
5th percentile	2.92	227	726	22.3	−6.33	28
95th percentile	11.15	664	1031	67.9	20.71	154
Matching criterion	1.55	84.9	61.1	9.21	3.53	33.8

Note: Temperature seasonality is the SD of average monthly temperatures and precipitation seasonality is the CV of average monthly precipitation.

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Our area of interest included drylands (defined as locations where the ratio of mean annual precipitation to mean annual potential evapotranspiration is <0.6 for 1970–2000) in the western United States and Canada (7.52×10^6 cells covering about 4.74×10^6 km^2; Appendix S4: Figure S1; Trabucco & Zomer, 2019). We conducted the simulations using current and future climate conditions at 41,477 dryland sites defined by a 10-km grid (detailed in Bradford et al., 2019). For each 10-km cell, we simulated conditions for five different soil types: site-specific soils derived from STATSGO (Miller & White, 1998) and four fixed soil types (Table 2), for a total of 207,385 sites with unique weather × soil combinations. Simulations included current conditions and two future time periods under two representative concentration pathways (RCPs) for all available general circulation models (GCMs) from CMIP5 (RCP4.5–37 GCMs; RCP8.5–35 GCMs; Taylor et al., 2012).

Inputs for SOILWAT2 include weather and climate data, soil information, and vegetation parameters. Site-specific vegetation parameters are estimated from climate (Bradford et al., 2014; Palmquist et al., 2016a), so we used two soil and six climate variables, which represent the major drivers of ecohydrology and plant community structure in drylands, as matching variables (Table 3). Because we simulated five soil types for each site, we tested several matching schemes to adapt the matching method to fit this experimental design (Appendix S5). We achieved the best matching using a two-step process. In the first step, target cells were matched to subset cells by finding the nearest neighbor using climate variables. In the second step, the nearest neighbor among the available soil types was determined (Table 2). We used DayMet (Thornton et al., 2018) data to calculate 30-year (1981–2010) normal monthly precipitation and temperature using Google Earth Engine (Gorelick et al., 2017) and obtained soil data from SoilGrids+ (Hengl et al., 2017; Appendix S4). We set the matching criterion to 5% of the range of each variable (a range that could be varied based on study-specific criteria) and further limited mean annual precipitation and mean annual temperature to 25 mm and 0.5°C, respectively (Table 3).

TABLE 2 Soil types used in the interpolation case study.

Soil type	Sand (%)	Clay (%)	Silt (%)
Clay loam	27	35	38
Sandy loam	66	9	25
Silt loam	16	9	75
Median[a] (silt loam)	30	18	52
Site specific	Variable	Variable	Variable

[a]Median soil type for all big sagebrush ecosystems (Palmquist et al., 2021; Appendix S4).

Evaluating matching and estimating matching error

We provide four metrics to quantify the quality of the match between simulated sites and target cells (detailed examples are provided in the interpolation vignettes in the R package). Importantly, the first three metrics are meant to evaluate the quality of matching and can be used to fine-tune matching criteria to balance matching uncertainty with the area that can be reasonably represented by the subset cells (Figure 2a). The fourth metric provides a means for estimating interpolation errors.

1. The site selection and interpolation methods match every target cell to a simulated site based on minimum Euclidean distance of the weighted matching variables. Thus, the first evaluation of matching is this distance, which can be interpreted as a continuous variable measuring matching quality (calculated by the "multivarmatch()" function). Distances less than or equal to one indicate "high-quality" matching (i.e., user-designated matching criteria have been met for all variables). Distances greater than one indicate that the difference between these target cells and their matched simulated site exceeds the matching criterion for one or more variables. This measure of matching quality can be used during interpolation to exclude target cells for which there is insufficient matching, that is, no simulated site is analogous, and to determine the spatial extent of the represented area. Importantly, the designation of distances of less than or equal to one as "high-quality matching" is not an objective measure of matching quality but instead depends on the choice of matching variables, matching criteria, and study system. Smaller values for matching criteria will reduce the uncertainty in interpolation and decrease the represented area. Larger values will increase the represented area but increase uncertainty (Figure 2a). Mapping matching quality in geographic space (e.g., Figure 3) can provide insight into spatial patterns of uncertainty in matching and interpolation.
2. Matching can be evaluated by calculating the SD of the pairwise differences between target and matched subset cells for a set of variables relevant to the project (using the "evaluateMatching()" function). This provides an estimate of how much variability there is among cells matched to the same site and is

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TABLE 3 Summary of matching variables for the interpolation case study with matching criteria (maximum difference between subset and target cells), defined as 5% of the range of each variable for all but mean annual precipitation and temperature (for which matching criteria were set to <5% of their range).

Metric	Mean annual temperature (°C)	Mean annual precipitation (mm)	Temperature seasonality	Precipitation seasonality	Mean temperature of driest quarter (°C)	Mean precipitation of warmest quarter (mm)	Sand (%)	Clay (%)
Mean	8.9	472.4	929.7	53.3	4.2	147.5	42.4	24
5th percentile	1.5	201.2	698.4	25.9	−12.2	20.9	24	13
95th percentile	18.4	870.1	1254.1	76.8	23.7	275.6	62.4	34.6
Matching criterion	0.5	25	62.8	5	2.5	19.9	4.4	3.8

Note: Temperature seasonality is the SD of average monthly temperatures and precipitation seasonality is the CV of average monthly precipitation.

particularly informative if the variables were not used as part of the matching process. If variability is too high, matching criteria can be adjusted to require stricter matching—though this will also reduce the area that can be represented by the target cells (Figure 2a).

3. The quality of matching will also depend on the extent to which spatiotemporal patterns are maintained between sites. Environmental and climatic variables generally exhibit spatial autocorrelation, and geographically adjacent sites should be more similar than geographically distant sites. Because our methods do not explicitly incorporate geographic proximity, our interpolation method could assign simulation output from geographically distant target cells to adjacent subset cells, potentially resulting in unrealistic spatial patterns in interpolated output. For example, in our case studies, sites from the eastern part of the region can be very similar to sites in the western part of the region in terms of mean annual precipitation but exhibit substantial differences in precipitation seasonality and vegetation type (Lauenroth et al., 2014). As a result, these sites could match based on annual climate variables, introducing spatial incoherence in variables at smaller temporal scales (e.g., monthly or daily values). Thus, geographic distance between target cells and their matched subset cells and between subset cells matched to adjacent target cells can be evaluated as proxy measures for spatial coherence in interpolated variables (i.e., interpolated variables will be more likely to exhibit spatial autocorrelation at all temporal scales, as expected). Two measures of distance can be calculated (using the "evaluateGeoDist()" function) to estimate the extent to which spatiotemporal patterns have been maintained during matching: (1) the geographic distance between target cells and their matched simulated sites, and (2) the average geographic distance between the simulated site that is matched to a given target cell and the simulated sites that are matched to the eight adjacent neighbors of that target cell. If these distances are large, it may be an indication that matching variables are not capturing regional differences in environmental variables, and we suggest trying additional or different matching variables. Importantly, in a study system where interpolated variables are not expected to exhibit spatial autocorrelation, these distance metrics may not be useful for evaluation.
4. Finally, once simulations are complete for the subset cells, interpolation errors can be estimated using leave-one-out cross validation (using the "loocv()" and "cverrors()" functions). To perform leave-one-out cross validation of interpolated variables, one simulated site is held out from the rest and matched to its nearest environmental neighbor from the remaining simulated sites using weighted matching variables. The interpolation errors for each interpolated variable are calculated as the difference between the simulation output from the held-out site and the simulation output from the matched site. This is repeated for all simulated sites, and an estimate of the interpolation error can be calculated for any interpolated output variable using the following equation:

$$\text{CVerror} = \sqrt{\frac{1}{k}\sum_{i=1}^{k}(y_i - \widehat{y_i})^2}, \quad (1)$$

where k is the number of simulated sites, y is the value of the simulated output variable at each held-out site, and $\widehat{y}$ is the value of the simulated output variable at the matched site. We calculated evaluation metrics 1–3 for the site selection case study and we calculated all four evaluation metrics for the interpolation case study.

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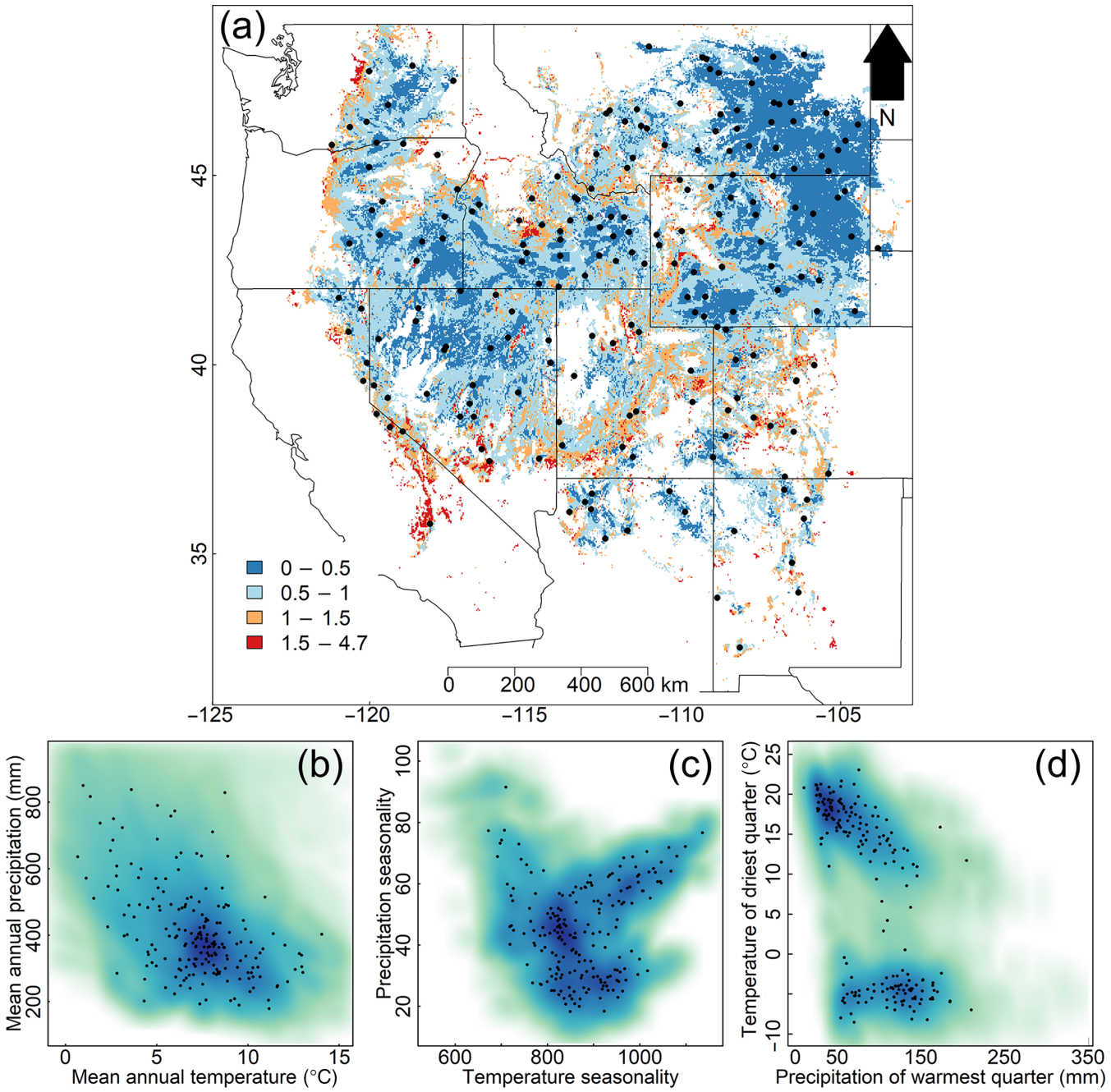


FIGURE 3 Map of the 200 subset cells (black points) that represent the big sagebrush region and matching quality (colors; Euclidean distance of the weighted matching variables) across this region (a). Below (b–d), the subset cells (black points) are shown in climate space of the six matching variables. The density of target cells in these panels is indicated by the background shading (darker colors indicate higher density), and the panels show all cells in the study area, including those with poor matching quality (weighted Euclidean distance >1.5). Temperature seasonality is the SD of average monthly temperatures and precipitation seasonality is the CV of average monthly precipitation.

RESULTS

Method 1: Selecting optimal sites for simulation and/or sampling

The proportion of the study area represented by the subset cells increased steadily up to approximately 200 sites, where incremental gains in area represented by additional sites began to decrease (Figure 2b). Thus, we selected 200 sites as the optimal number that could reasonably represent our region while minimizing computational burden. Using the 200 sites, we achieved high-quality matching (evaluation metric 1: weighted Euclidean distance ≤1) for about 93% of the study area. We achieved

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moderate matching quality (weighted Euclidean distance between 1 and 1.5) for an additional 3% of the study area (Figure 3a). We used all cells with high- and moderate-quality matching to define our final study area, achieving 96% coverage of the original area of interest (1.15×10^6 km^2 represented by 1.82×10^6 cells). The sites were well distributed across the geographic and climate space of the region (Figure 3).

Cells with poor quality matching were found in topographically rich areas, at the edges of the study area, and in areas with sparse or isolated cells (Figure 3a). Of the additional climate variables that we examined to further evaluate matching (evaluation metric 2), most temperature variables had relatively low SD of pairwise differences between matched subset and target cells (<2°C; Appendix S6: Figure S5) and spatial patterns of differences were similar to those of the matching variables (Appendix S6: Figures S1–S3). The mean temperature of the wettest quarter of the year was an exception. The SD of differences between target and subset cells was ~3.5°C, and we found discrepancies of over 15°C along the western side of the Colorado Plateau and in central Idaho and northern Washington. For the precipitation variables that we examined, the SD of differences between target and subset cells was relatively low (<15 mm) (Appendix S6: Figures S4 and S5), and the largest differences were found in topographically rich areas (Appendix S6: Figure S4).

The median geographic distance between simulated sites and target cells (evaluation metric 3a) was 81 km and ranged from 0 to 1724 km. Larger distances were found in topographically rich areas, west central and southeast Wyoming, eastern Washington, along the Snake River in southwestern Idaho, and in eastern Oregon (Appendix S6: Figure S6). The median of the average distance between the simulated site matched to a target cell and the simulated sites matched to the eight adjacent neighbors (evaluation metric 3b) was 0 km and ranged from 0 to 1309 km (Appendix S6: Figure S7).

Method 2: Interpolation

We achieved high-quality matching (evaluation metric 1) for 91% of western North American drylands for climate matching variables and for 19% of the area for soil matching variables (Appendix S7: Figures S1 and S2). We included all target cells with a climate matching quality value of ≤1.5 in our final interpolated area, resulting in 92% coverage of North American drylands (4.38×10^6 km^2 represented by 6.92×10^6 cells). Areas with poor matching were mostly found in regions that were classified as drylands at 30-arcsecond resolution but not at 10-km resolution in the Rocky Mountains, adjacent to the Sierra and Cascade Mountain ranges, and along the eastern and northern edges of the study area (Appendix S4: Figure S1 and Appendix S7: Figures S1, S5, S6). We did not exclude any cells with poor matching quality of soil variables because soils can be highly variable at scales smaller than our interpolation efforts and because climate has a greater influence than soils over ecohydrological conditions in these ecosystems (Lauenroth et al., 2014). Soil information from the simulated site matched to each target cell is shown in Appendix S7: Figures S3 and S4.

Of the additional climate variables that we examined to further evaluate matching (evaluation metric 2), the SDs of differences between target and simulated sites were low (<0.5°C) for most of the temperature variables and did not exhibit distinct geographical patterns (Appendix S7: Figures S7, S8, S10). However, the SD of differences was 1.6°C for mean temperature of the wettest quarter, with larger differences found in topographically rich areas in the western part of the study area (Appendix S7: Figure S8). All precipitation variables that we examined had low SDs of differences (<5 mm) between target and subset cells, and larger differences were not concentrated geographically (Appendix S7: Figures S9 and S10).

The median geographic distance between simulated sites and target cells (evaluation metric 3a) was 12 km and ranged from 0 to 2106 km. Larger distances were found in topographically rich areas and along the eastern and northern edges of the study area (Appendix S7: Figure S11). The median of the average distance between the simulated site matched to a target cell and the simulated sites matched to the eight adjacent neighbors (evaluation metric 3b) was 6 km and ranged from 0 to 2137 km (Appendix S7: Figure S12).

Finally, we estimated interpolation errors using leave-one-out cross validation (evaluation metric 4; Appendix S8) for six output variables presented in Bradford et al. (2019). Our estimated interpolation errors were relatively small (CV_{error} < 10% of the range of each variable). The differences between simulated and matched output variables were approximately normally distributed, with means very close to 0 (Figure 4). Although differences appeared to be larger in topographically rich areas, there were no distinct geographic patterns in these differences (Appendix S8: Figure S4). The cross-validated matching resulted in correctly assigned soil temperature and moisture (STM) regimes (one of the categorical output variables of Bradford et al., 2019) for 66% of cells (i.e., simulated STM = matched STM). STM regimes were incorrectly assigned to 4% of cells and the remaining 29% of cells were assigned correctly to either soil

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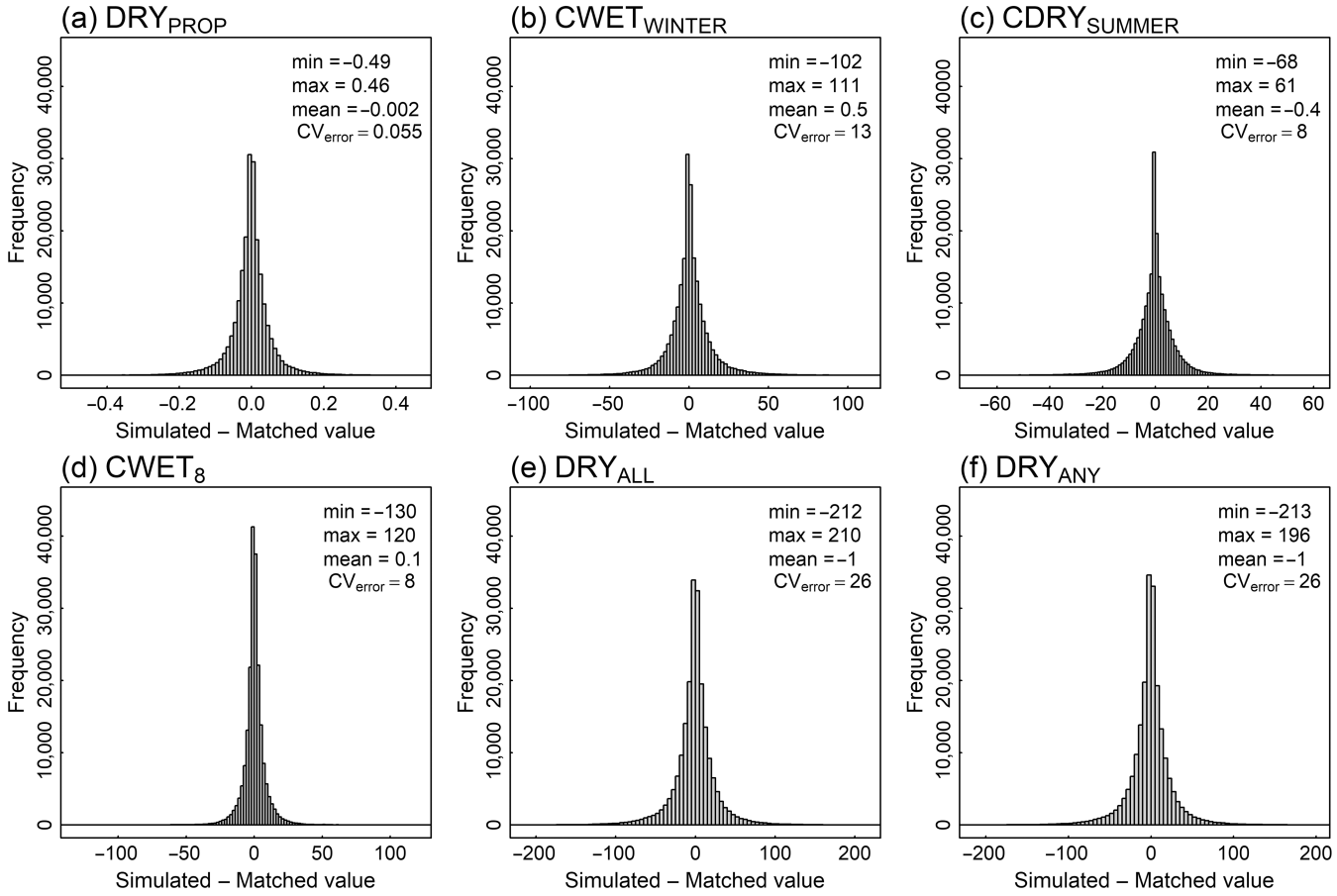


FIGURE 4 Estimated interpolation errors for six output variables from a study investigating soil temperature and moisture regimes in western North America (Bradford et al., 2019). These variables describe the frequency and seasonality of wet (>−1.5 MPa) soil conditions within the moisture control section (MCS: soil layers with depth ranging from 10–30 cm for fine textures to 30–90 cm for coarse textures; Soil Survey Staff, 2014). DRY_{PROP} (a) refers to the proportion of days that all layers within the MCS are dry when soil temperature at 50 cm is >5°C, $CWET_{WINTER}$ (b) is the number of consecutive days with all MCS layers wet during the winter, $CDRY_{SUMMER}$ (c) is the number of consecutive days with all MCS layers dry during the summer, $CWET_8$ (d) is the number of consecutive days with any layer wet when soil temperature at 50-cm depth is >8°C, DRY_{ALL} (e) is the number of days with all MCS layers dry, and DRY_{ANY} (f) is the number of days when any soil layer in the MCS is dry.

temperature or soil moisture regime, but not to both. The results of this matching and interpolation method are reasonably accurate continuous maps of SOILWAT2 output variables across western North America that provide greater detail than maps generated by the coarse grid of points used to define simulations (Figure 5). Furthermore, although high-resolution maps of single variables are useful, our method, by design, interpolates highly multivariate and time series data (e.g., daily soil water availability at multiple soil depths, Figure 6).

DISCUSSION

The spatial resolution of information generated by complex ecological simulation models is often limited by computational restrictions and thus it can be a challenge to represent responses to fine-scale heterogeneity in topography, soils, and other environmental variables. We addressed this limitation by first developing a method to select a minimum number of sites from a given study area that maximize the area represented by those sites (Method 1). Then, we developed a multivariate interpolation method that matches simulated sites to all locations across the study area to produce high-resolution, continuous maps of simulation output (Method 2), with estimates of interpolation error. Our methods are designed to facilitate the application of sparse simulation model results to local-scale resource management and conservation decisions, which often require high-resolution information about complex ecological processes in heterogeneous regions.

We demonstrated these methods via two case studies using results from our previous work projecting climate

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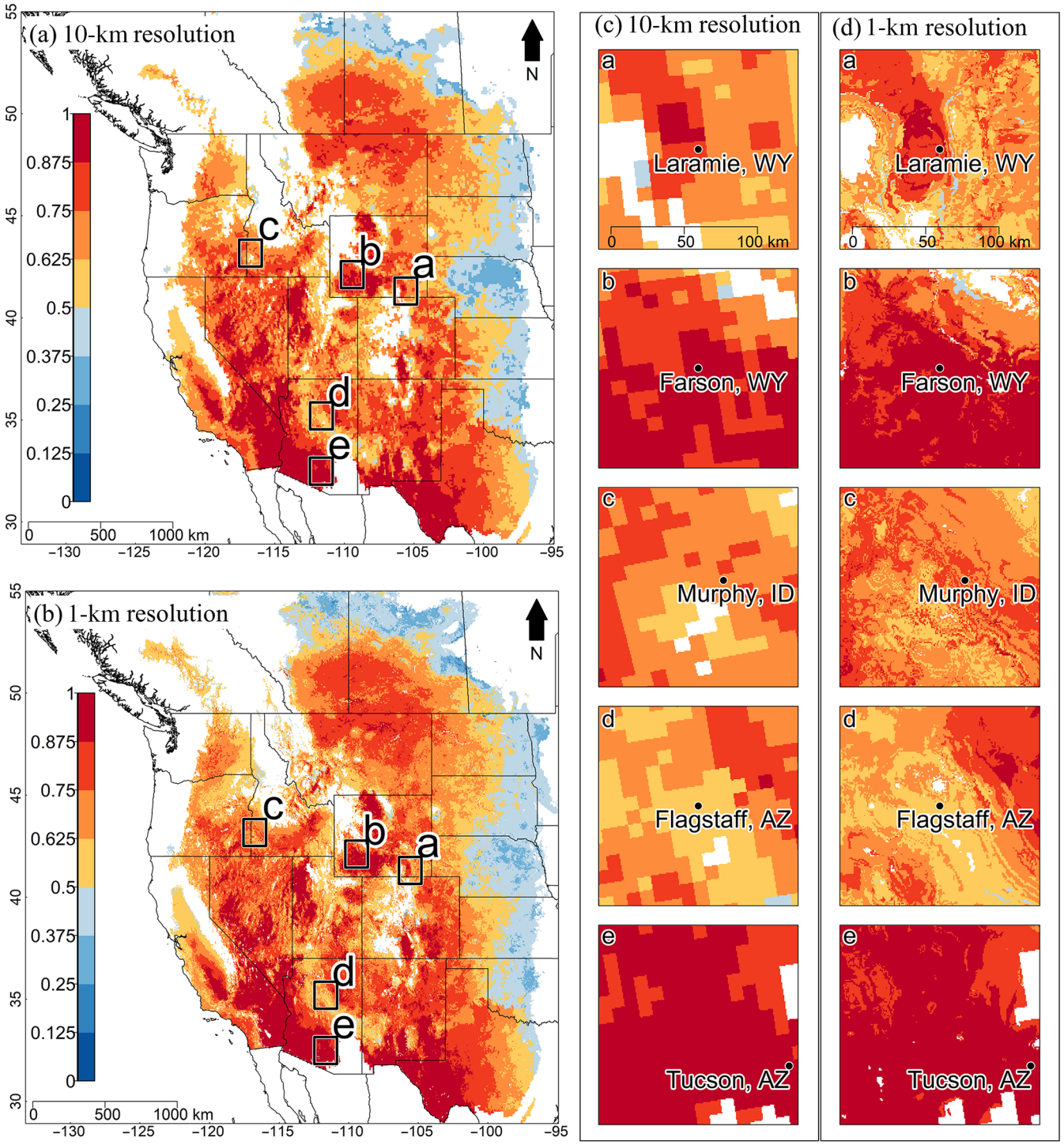


FIGURE 5 Maps of simulation results for site-specific soils at points representing a grid with 10-km resolution across North American drylands (a) and high-resolution (30-arcsecond) interpolated results (b) for DRY_{PROP}, the proportion of days in which all layers within the moisture control section (soil layers with depth ranging from 10–30 cm for fine textures to 30–90 cm for coarse textures; Soil Survey Staff, 2014) of the soil are dry when soil temperature at 50 cm is >5°C (Bradford et al., 2019). Panels (c) and (d) provide detailed maps corresponding to the inset boxes in (a) and (b).

change and disturbance impacts on big sagebrush plant communities in the western United States (case study 1; Palmquist et al., 2021), and investigating current and future STM regimes in North American drylands (case study 2; Bradford et al., 2019). The 30-arcsecond (~1 km) resolution maps we produced can resolve spatial patterns across management areas (e.g., Figure 6), with application to management decisions at local and regional

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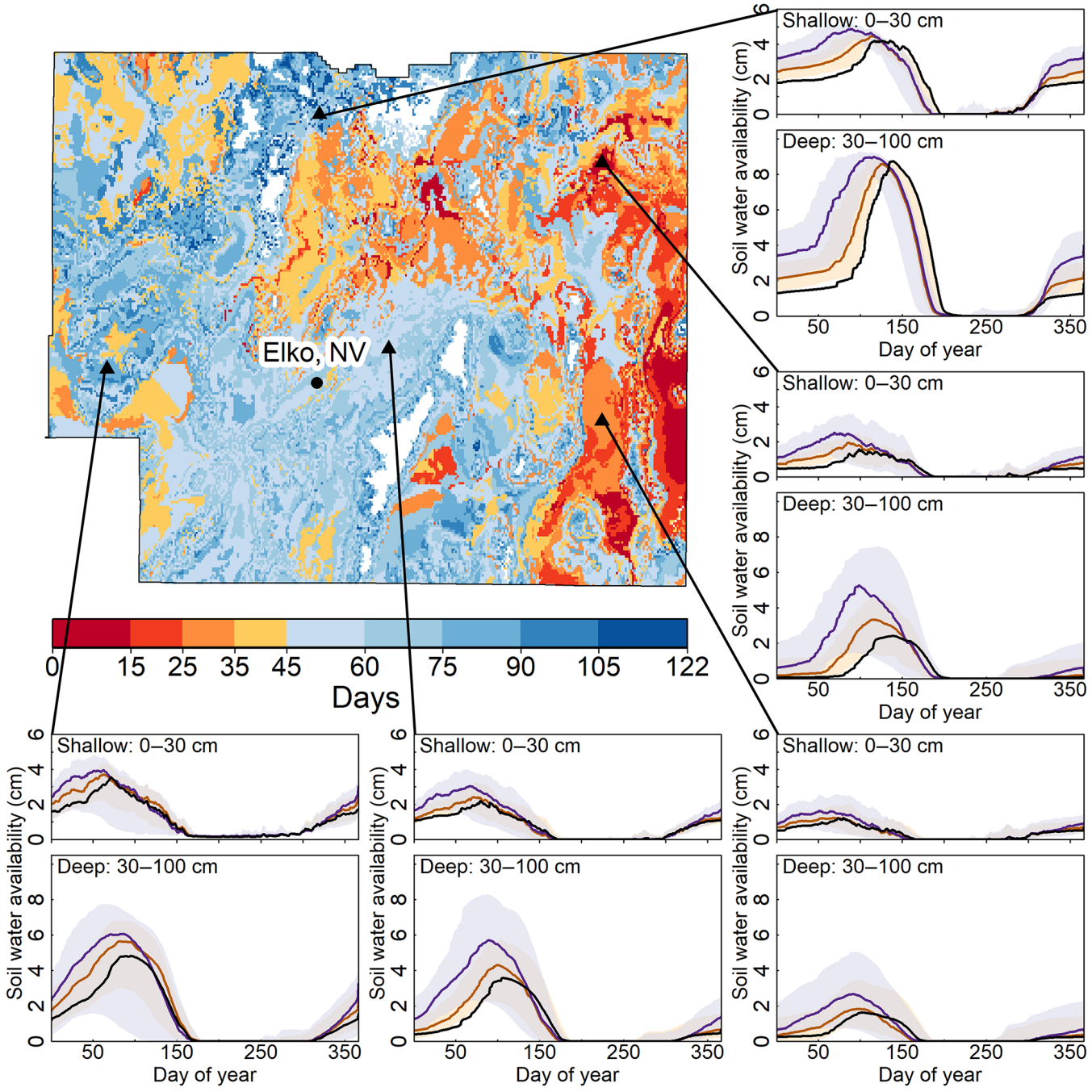


FIGURE 6 Interpolated map of $CWET_{WINTER}$ for the Elko Bureau of Land Management District in northeastern Nevada. $CWET_{WINTER}$ is the number of consecutive days with all moisture control section (soil layers with depth ranging from 10 to 30 cm for fine textures to 30–90 cm for coarse textures; Soil Survey Staff, 2014) soil layers wet during the winter, Bradford et al. (2019). Panels show average soil water availability (SWA) under current (black), mid-century (2020–2050) RCP8.5 (gold), and end of century (2070–2100) RCP8.5 (purple) conditions for five locations. Lines depicting future SWA represent the median and shaded areas show the range for each day across 35 general circulation models. RCP, representative concentration pathway.

scales. Although our interpolated maps are a substantial improvement over coarser-scale or geographically sparse simulations, soils and vegetation can vary at scales smaller than the resolution of our interpolated variables here. Our methods could produce higher resolution maps if matching variables were available at such a resolution.

Our methods rely on a relatively small subset of variables to match simulated sites to target cells. We simulated ecosystem water balance and plant community composition in drylands and chose climate (or climate and soil) variables that are the major drivers of soil water dynamics and plant community structure in North American drylands. We used six to eight variables for

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each case study and assessed matching success by confirming that sites were also relatively similar in terms of other climate variables (Appendix S6: Figures S2–S5 and Appendix S7: Figures S7–S10). The application of these methods for other studies will require selection of matching variables that are relevant to the modeling approach and the study system. The number of matching variables could be larger than what we used in these demonstrations, but increasing the number of variables will make the approach more computationally intensive.

In our interpolation case study, the experimental design of the simulations meant that we needed to match sites based on both climate and soil texture. Our efforts to match in a single step that incorporated both climate and soils resulted in large differences in climate variables (Appendix S5), and we were challenged to determine matching criteria that would be equivalent for climate and soils. Thus, we adapted our matching method to fit the needs of our study and completed matching as a two-step process. We suggest that future use of our methods carefully consider experimental design to determine an appropriate matching scheme. Our R package can be used for two-step matching in cases where experimental design is similar to ours using the "multivarmatch()" and "secondaryMatching()" functions (see the *Two-step matching and interpolation without kpoints* vignette for details).

Our method of interpolation uses multivariate matching to determine climatic and environmental analogs for all cells in an area of interest from a subset of simulated sites, and then assigns the complete output of results from each simulated site to all the analogous cells. Thus, the method is able to interpolate highly multivariate and time series data and maintains internal consistency in the structure and distribution of output variables for a given site, avoiding the need to correct interpolated variables to maintain realistic values (e.g., Hengl et al., 2017). Our method is cost-effective and provides computational advantage over more complex emulator methods (we performed matching for millions of cells on a laptop). It is flexible and uses environmental analogs to overcome geostatistical challenges of interpolating sparse multivariate data in complex terrain. Furthermore, we have developed our methods into an R package with vignettes that provide detailed step-by-step instructions.

In both case studies, our methods successfully found environmentally analogous simulated sites for most target cells. However, the larger discrepancies in temperature of the wettest quarter suggest that simulated sites may not have adequately matched precipitation seasonality in some locations. Adding additional matching variables could improve the representation of this variable. For the interpolation case study, soil matching quality was low and only 19% of the study area had matching quality ≤ 1. This may be in part due to differences in the data sources used to derive soils for subset and target cells. The soils used for the simulations were, on average, lower in sand and clay content than those derived from SoilGrids+ (Hengl et al., 2017) for the interpolated cells (Appendix S7: Figures S3 and S4).

Importantly, our methods do not explicitly incorporate geographic proximity, which could result in inconsistencies in spatiotemporal patterns between target cells matched to geographically distant simulated sites. In our case studies, the small median distance between subset and target cells and adjacent matched cells suggests that matching maintained spatial coherence relatively well. However, this issue could be addressed in future implementations by incorporating geographical distance into the algorithms, (e.g., matching target cells to the nearest geographical neighbor from all subset cells within a given weighted Euclidean distance of the matching variables and/or specifying a maximum allowable geographic distance).

Our site selection method is highly flexible and allows users to balance the cost of simulating additional sites with the benefit of enlarging the area represented by those sites. Although other point selection techniques (e.g., conditioned Latin hypercube (Minasny & McBratney, 2006) or systematic spatial point selection) do not consider the relative importance or scale of variables used in matching, our method specifically integrates user-specified criteria for the maximum allowable difference for each matching variable. This allows users to leverage domain expertise to choose matching variables and define matching criteria, then use the evaluation metrics to fine-tune matching criteria to balance the matching uncertainty with the size of the study area that can be represented by the subset cells (Figure 2a). Furthermore, this method has potential to improve the efficiency of field-based observational studies by finding an optimal number of field site locations necessary to adequately represent a study area. In addition, the interpolation method presented here is sufficiently simple from a computational standpoint that it provides an obvious advantage in situations where computational capacity would limit both the number of simulated sites and implementation of an algorithm that would be capable of interpolating complex simulation output. Together, our multivariate matching algorithms and the rMultivariateMatching R package provide a flexible and relatively simple method of efficiently leveraging computational resources to produce continuous and geographically extensive estimates of multivariate ecological variables at scales relevant to local and regional management decisions, with clear paths for evaluating matching accuracy and interpolation error.

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AUTHOR CONTRIBUTIONS

The project was conceived collectively by the authors. Kyle A. Palmquist ran the STEPWAT2 simulations and Daniel R. Schlaepfer ran the SOILWAT2 simulations. Rachel R. Renne developed the methods and code base, performed the analyses, and wrote the manuscript with contributions from John B. Bradford, Daniel R. Schlaepfer, Kyle A. Palmquist, and William K. Lauenroth.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data produced using methods described here (Renne et al., 2021, 2022) are available from the U.S. Geological Survey's ScienceBase-Catalog: https://doi.org/10.5066/P9MNKWS4, https://doi.org/10.5066/P950E9GH. Code (Renne, 2021) is available from Zenodo: https://doi.org/10.5281/zenodo.5784247.

ORCID

Rachel R. Renne https://orcid.org/0000-0002-6798-9533
Daniel R. Schlaepfer https://orcid.org/0000-0001-9973-2065
Kyle A. Palmquist https://orcid.org/0000-0001-7665-4105
John B. Bradford https://orcid.org/0000-0001-9257-6303

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SUPPORTING INFORMATION

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